SOFTWARE ENGINEERING

SMART CARE SYSTEM

Student Name and IDS

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The smart Care System is an application built for medical clinics that are struggling with synchronizing patient records and scheduling appointments while ensuring efficiency. The smart care System is designed to specifically replace faulty unreliable systems while ensuring quality patient care. This is important because it will empower the health sector to operate smoothly by relieving it of its administrative burdens allowing medical workers to focus on what matters the most which is the patients.

Scrum Framework Application

The scrum was applied in terms of the product backlog by creating 8 different PBIs which included user registration, booking and canceling appointments, sending reminders.

Sprint planning with the sprint planning we went ahead to make 3 user stories for week 2 sprint

For the sprint backlog we selected User Registration and Login, book appointments, view available slots

With all this through constant meetings we managed to improve on our weaknesses by identifying what went wrong and improving on it

The Roles were assigned as follows

Scrum Master- Timothy Semafumu

Development Team – Kamlesh Le Hoang and Timothy.

Team Roles and Responsibility

ROLE TEAM LEADER – TIMOTHY SEMAFUMU MANAGED THE PRODUCT BACKLOG AND PRIORITIZED THE FEATURES . ALSO AS A SCRUM MASTER ENSURED ALL SCRUM PROCESSES WERE FOLLOWED

DEVELOPER AUTHENTICATION – LE HOANG USER REGISTRATION, LOGIN ROLE-BASED ACCESS DATABASE SETUP ALSO DEVELOPED THE BOOKING SCHEDULING

DASHBOARD DEVELOPER BY KAMLESH DOCTOR DASHBOARDS PATIENTS DASHBOARD

TESTING WAS CARRIED OUT BY ALL STUDENTS USING UNIT TESTING, BLACK BOX TESTING WHITE-BOX TESTING

Smart care system users

- Administrator
- Receptionist
- Doctor
- Patients

Technologies used

Frontend: HTML/CSS

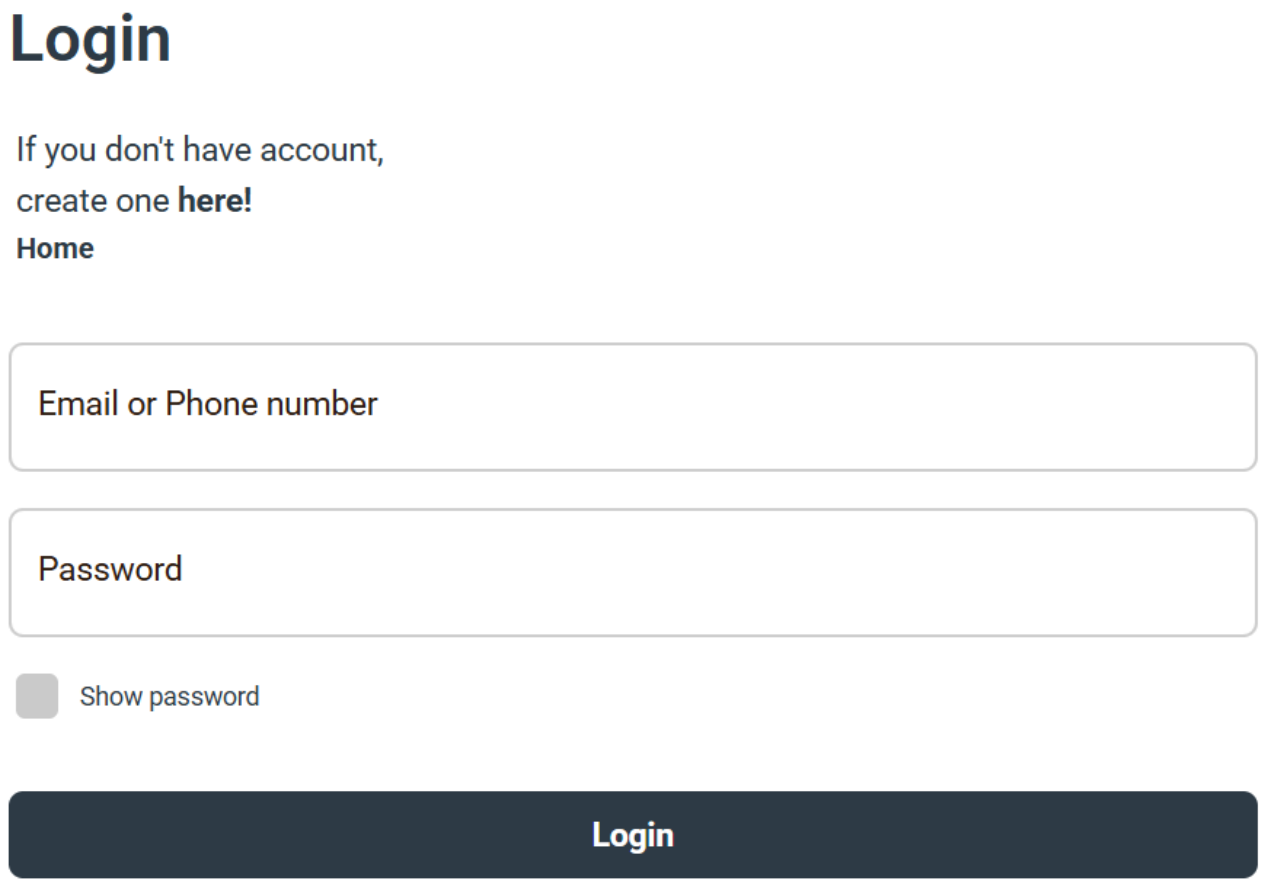
Backend: Node.js

Database: MySQL (for better visualization we used prisma)

Security : Authentication Separation of Duties(SOD)

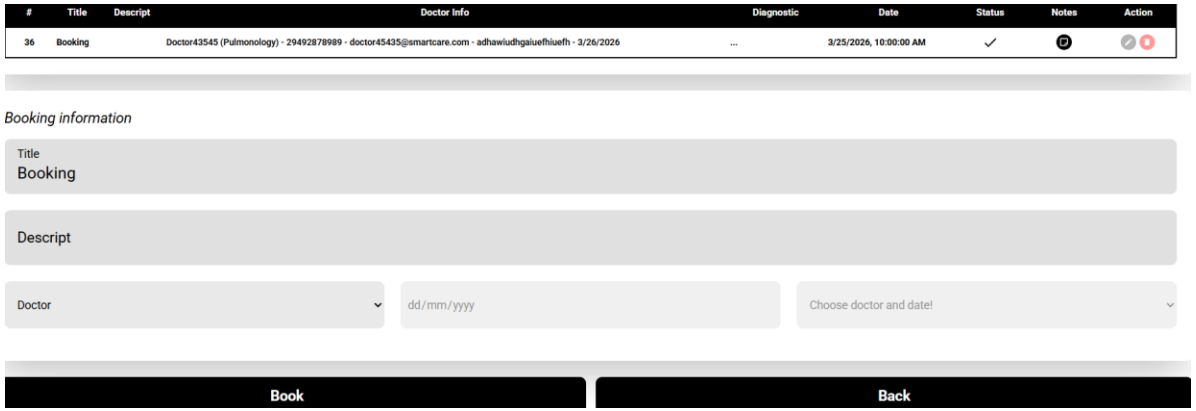
Implemented core use cases

- User registration and login



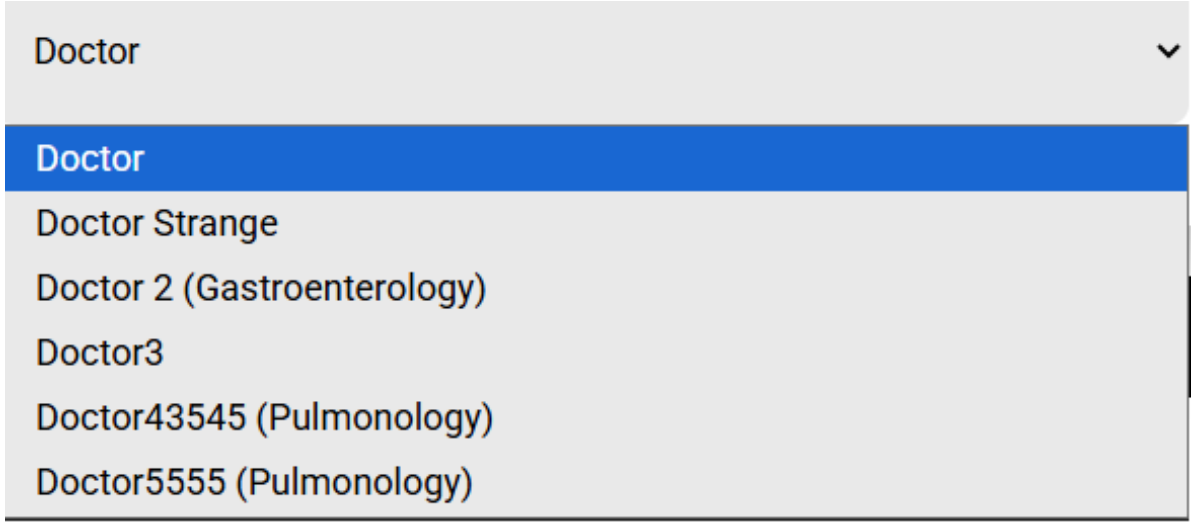
Implemented core use cases

- Book Appointment



Implemented core use cases

- View available doctors



Challenges encountered

- When multiple users tried to book the same slot for an appointment this succeeded which will lead to conflicts therefore implemented a database transaction locking to ensure there's no double booking hence eliminating it

The Road Ahead

- Patient self-booking portal.
- Email/SMS reminders for appointments.
- Admin analytics dashboard (patient volume).
- Final security audit and testing.

Thank You

QUESTIONS ?

HTTPS://GITHUB.COM/ANDREWSEMAFUMU/SOFWAREENGINEERING